



Site



APT-29 LOTL

Adversary Emulation

Active Directory • Threat Hunting • Incident Response

A safe, isolated lab that reproduces a realistic attack chain — then detects, investigates, and responds to it using Elastic Security.

Team members

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Why this project?

The Problem

SOC teams often can't catch nation-state intrusions until damage is done, because attackers abuse trusted admin tools such as powershell instead of malware.

Detection Gap: without behavior-based visibility, Living-off-the-land (LOTL) techniques can evade or reduce the effectiveness of traditional signature-based detection

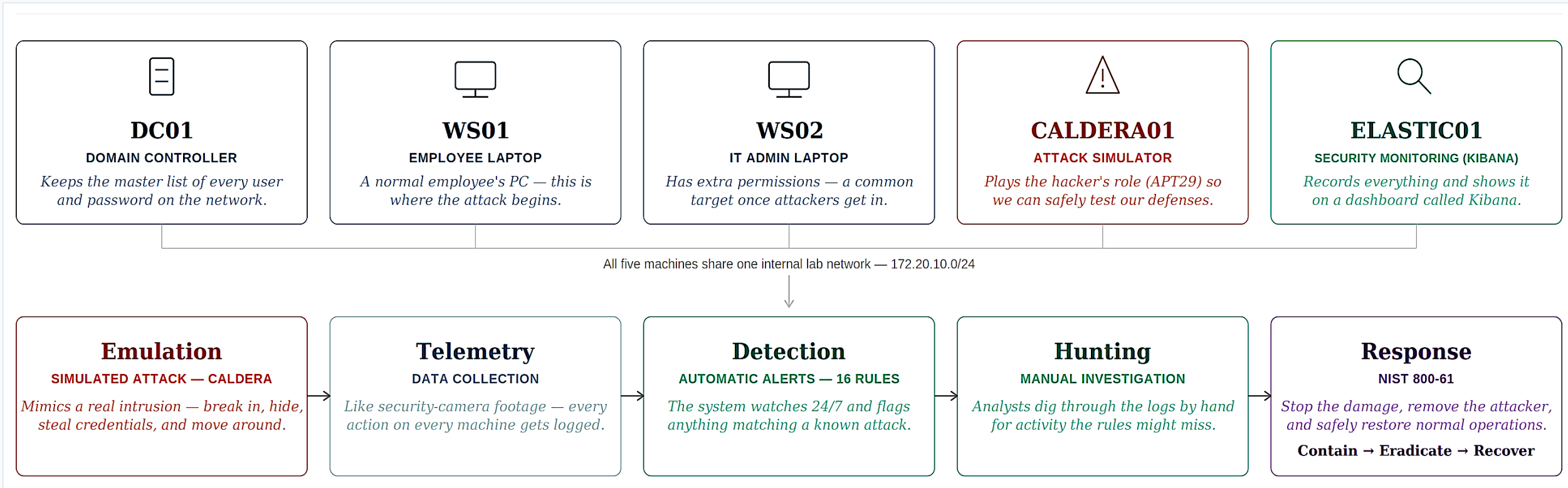
Our Approach

Build a realistic Active Directory lab, safely emulate an APT29-inspired attack chain, collect high-fidelity telemetry, and engineer custom behavioral detections.

Why adversary emulation is useful:

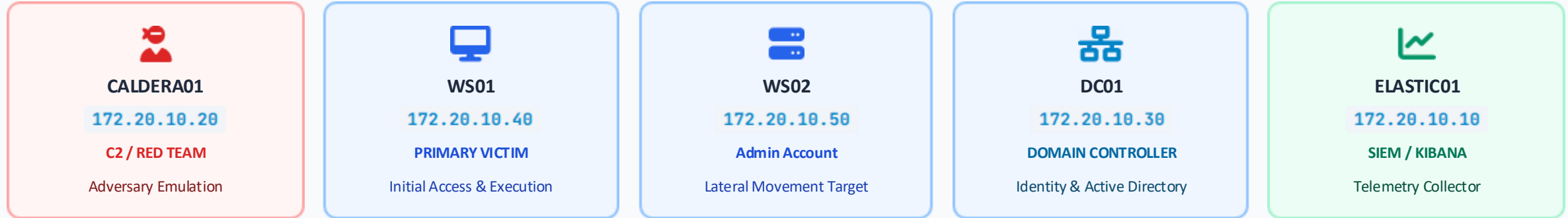
- **For analysts** — realistic reps against real TTPs, not tabletop theory
- **For detection engineers** — rules validated against actual attack behavior, not assumptions
- **For leadership** — measurable readiness.

Project Architecture



Environment

Five isolated hosts operated on the **172.20.10.0/24** lab subnet. DC01, WS01, and WS02 belonged to the **corp.local** domain, while CALDERA01 and ES01 operated as standalone attack-emulation and monitoring platforms.



Attack Emulation Platform

Automated APT29 Living of the land technique execution via MITRE CALDERA framework targeting Windows workstations and Active Directory components.

Telemetry Sources

Comprehensive logging pipeline: **Sysmon** + **Windows Event Logs** + **PowerShell Script Block Logging**

Attack Chain

Executed 3 CALDERA adversary profiles alongside honeypot account interaction across 8 distinct phases:

1

Initial Access

mshta → PowerShell

2

Execution

Obfuscated PowerShell

3

Discovery

System + AD Recon

4

Persistence

Run Key + Task + WMI

5

Lateral Movement

WinRM → WS02

6

Credential Access

DCSync / Mimikatz

7

Collection

CompanyData Staging

8

Exfiltration

HTTP → :9999

MITRE ATT&CK® Mapping Coverage

T1218.005 (Mshta)

T1059.001 (PowerShell)

T1547.001 (Registry Run Keys)

T1053.005 (Scheduled Task)

T1021.006 (WinRM)

T1546.003 (WMI Event)

T1003.006 (DCSync)

T1119 (Automated Collection)

T1041 (Exfiltration Over C2)

Elastic Detection Rule Development

Elastic Settings:

16

Detection Rules

1 min

EXECUTION INTERVAL

5 min

LOOK-BACK WINDOW

KQL / EQL

QUERY LANGUAGES

Detection Categories:



Process Behavior

Parent/child anomalies, encoded PowerShell payloads, and suspicious execution paths (e.g., mshta spawning powershell).



Persistence & Identity

Registry Run key additions, scheduled task registrations, WMI subscriptions, WinRM sessions, and DCSync replication.



Network & Collection

Staging of company data, suspicious outbound PowerShell HTTP traffic, and C2 exfiltration on non-standard ports (:9999).



Key Implementation Insight: PowerShell Script Block content was validated specifically in **powershell.file.script_block_text** — rather than the default field initially anticipated during early engineering tests.

Detection Results

After two complete attack-chain executions, the optimized final run produced reliable telemetry and high-confidence alerts.

39

FINAL-RUN ALERTS

8

ATTACK PHASES COVERED

3

Rules TUNED

DCSync

Captured

Validation Detection Results

- **DCSync:** Elastic detected the simulated DCSync activity after Directory Service Access auditing was enabled on DC01.
- **Encoded PowerShell:** Event ID 4104 exposed the decoded script content despite Base64 command-line encoding.
- **Honeypot Activity:** Interaction with the decoy account generated an immediate security alert.

Lessons and Improvements

- **Rule tuning:** Three detection rules were adjusted to reduce false-positive alerts during repeated testing.
- **WinRM visibility:** Although one CALDERA ability encountered an execution issue, Elastic still captured the associated WinRM authentication and PowerShell telemetry.
- **Telemetry correction:** Detection queries were updated to use the PowerShell field actually populated in Elastic.

Threat Hunting: Connecting the Attack Chain

Six targeted hunts correlated endpoint, identity, persistence, and network activity across the environment.

Hunt ID	Investigation Focus	Artifact / Indicator	Target Host / Log
H1	CALDERA Child Processes	CALDERA Sandcat agent process disguised as splunkd.exe	Sysmon Event ID 1
H2	Registry Persistence	Run key creation / modification	Sysmon Event ID 13
H3	Scheduled Tasks	schtasks / Register-ScheduledTask	Security 4698 / Sysmon 1
H4	WMI Persistence	Event Filter / Consumer / Binding	Sysmon Events 19, 20, 21
H5	Command & Control Communications	Outbound HTTP → 172.20.10.20	Sysmon Event ID 3
H6	DCSync Replication	DS-Replication-Get-Changes GUIDs	Security Event ID 4662 (DC01)



Hunting outcome: The six investigations connected isolated events into a clear sequence from initial execution and persistence to command-and-control and DCSync activity.

Incident Response and Remediation

A structured NIST response process was used to investigate, contain, and remediate the simulated APT29 intrusion.

Step 1

Detect & Triage

Validate alert context, aggregate correlation logs in Kibana, confirm severity.

Step 2

Contain

Isolate compromised host WS01 from network; disable compromised AD user accounts.

Step 3

Eradicate

Remove Registry Run keys, scheduled tasks, WMI subscriptions, Mimikatz & CALDERA agent.

Step 4

Recover

Reset Kerberos Krbtgt & user credentials, re-enable network interfaces, restore host.

Step 5

Improve

Document lessons learned, tune detection thresholds, harden AD audit policies.

Manual Remediation Commands Executed

- Host isolation
- Account containment
- Persistence removal
- CALDERA agent removal
- System validation
- Detection-rule tuning

Emulation in Action

Live Alert Feed — Elastic Security

Actions	@timestamp	Rule	A...	Seve...	R...	Reason	host.name	user.name	process.name	file.n...	source.ip	desti...
   	Aug 5, 2026 @ 18:32:06.523	APT29 - Honeypot Account Access Detected	critical	99		authentication event with source 172.20.10.40:5...	dc01	svc-backup	—	—	172.20.10.40	—
   	Aug 5, 2026 @ 18:32:06.513	APT29 - System and AD Discovery	medi...	47		process event with process net.exe, parent proce...	ws01	ws01	net.exe	—	—	—
   	Aug 5, 2026 @ 18:32:06.505	APT29 - System and AD Discovery	medi...	47		process event with process net.exe, parent proce...	ws01	ws01	net.exe	—	—	—
   	Aug 5, 2026 @ 18:30:56.351	APT29 - WMI Event Subscription Persistence	critical	99		process event by Administrator on ws02 created ...	ws02	Administrator	—	—	—	—
   	Aug 5, 2026 @ 18:30:48.438	APT29 - PowerShell Script Block Suspicious Content	high	73		process event by Administrator on ws02 created ...	ws02	Administrator	—	—	—	—
   	Aug 5, 2026 @ 18:30:48.429	APT29 - PowerShell Script Block Suspicious Content	high	73		process event by Administrator on ws02 created ...	ws02	Administrator	—	—	—	—
   	Aug 5, 2026 @ 18:28:56.602	APT29 - WMI Event Subscription Persistence	critical	99		process event by Administrator on ws02 created ...	ws02	Administrator	—	—	—	—
   	Aug 5, 2026 @ 18:24:48.476	APT29 - PowerShell Script Block Suspicious Content	high	73		process event by Administrator on ws02 created ...	ws02	Administrator	—	—	—	—
   	Aug 5, 2026 @ 18:24:48.467	APT29 - PowerShell Script Block Suspicious Content	high	73		process event by Administrator on ws02 created ...	ws02	Administrator	—	—	—	—
   	Aug 5, 2026 @ 18:23:09.461	APT29 - Mimikatz Process Execution	critical	99		process event with process mimikatz.exe, parent ...	dc01	Administrator	mimikatz.exe	—	—	—
   	Aug 5, 2026 @ 18:23:09.451	APT29 - Mimikatz Process Execution	critical	99		process event with process cmd.exe, parent proc...	dc01	Administrator	cmd.exe	—	—	—
   	Aug 5, 2026 @ 18:23:07.908	APT29 - Process Execution from Suspicious Directory	medi...	47		process event with process mimikatz.exe, parent ...	dc01	Administrator	mimikatz.exe	—	—	—
   	Aug 5, 2026 @ 18:22:49.911	APT29 - DCSync Credential Dumping	critical	99		iam, configuration event by Administrator on dc0...	dc01	Administrator	—	—	—	—
   	Aug 5, 2026 @ 18:22:49.904	APT29 - DCSync Credential Dumping	critical	99		iam, configuration event by Administrator on dc0...	dc01	Administrator	—	—	—	—
   	Aug 5, 2026 @ 18:22:49.896	APT29 - DCSync Credential Dumping	critical	99		iam, configuration event by Administrator on dc0...	dc01	Administrator	—	—	—	—
   	Aug 5, 2026 @ 18:22:49.891	APT29 - DCSync Credential Dumping	critical	99		iam, configuration event by Administrator on dc0...	dc01	Administrator	—	—	—	—
   	Aug 5, 2026 @ 18:21:53.493	APT29 - PowerShell Suspicious Network Outbound	critical	99		process event with process powershell.exe, pare...	ws01	ws01	powershell.exe	—	—	—
   	Aug 5, 2026 @ 18:21:06.747	APT29 - File Collection via PowerShell	medi...	50		process event with process powershell.exe, pare...	ws01	ws01	powershell.exe	—	—	—
   	Aug 5, 2026 @ 18:21:06.743	APT29 - File Collection via PowerShell	medi...	50		process event with process powershell.exe, pare...	ws01	ws01	powershell.exe	—	—	—
   	Aug 5, 2026 @ 18:20:50.028	APT29 - PowerShell Script Block Suspicious Content	high	73		process event by ws01 on ws01 created high ale...	ws01	ws01	—	—	—	—
   	Aug 5, 2026 @ 18:20:49.966	APT29 - Explicit Credential Object Creation	high	73		process event by ws01 on ws01 created high ale...	ws01	ws01	—	—	—	—
   	Aug 5, 2026 @ 18:20:07.362	APT29 - File Collection via PowerShell	medi...	50		process event with process powershell.exe, pare...	ws01	ws01	powershell.exe	—	—	—
   	Aug 5, 2026 @ 18:19:50.136	APT29 - PowerShell Script Block Suspicious Content	high	73		process event by ws01 on ws01 created high ale...	ws01	ws01	—	—	—	—
   	Aug 5, 2026 @ 18:19:50.132	APT29 - PowerShell Script Block Suspicious Content	high	73		process event by ws01 on ws01 created high ale...	ws01	ws01	—	—	—	—
   	Aug 5, 2026 @ 18:19:50.127	APT29 - PowerShell Script Block Suspicious Content	high	73		process event by ws01 on ws01 created high ale...	ws01	ws01	—	—	—	—
   	Aug 5, 2026 @ 18:19:50.119	APT29 - PowerShell Script Block Suspicious Content	high	73		process event by ws01 on ws01 created high ale...	ws01	ws01	—	—	—	—
   	Aug 5, 2026 @ 18:19:49.528	APT29 - Scheduled Task Persistence	high	73		process event with process powershell.exe, pare...	ws01	ws01	powershell.exe	—	—	—
   	Aug 5, 2026 @ 18:19:06.639	APT29 - System and AD Discovery	medi...	47		process event with process reg.exe, parent proce...	ws01	ws01	reg.exe	—	—	—
   	Aug 5, 2026 @ 18:18:50.041	APT29 - Registry Run Key Persistence	high	77		configuration, registry event with process reg.exe...	ws01	ws01	reg.exe	—	—	—
   	Aug 5, 2026 @ 18:18:06.824	APT29 - System and AD Discovery	medi...	47		process event with process net.exe, parent proce...	ws01	ws01	net.exe	—	—	—
   	Aug 5, 2026 @ 18:18:06.819	APT29 - System and AD Discovery	medi...	47		process event with process net.exe, parent proce...	ws01	ws01	net.exe	—	—	—
   	Aug 5, 2026 @ 18:18:06.815	APT29 - System and AD Discovery	medi...	47		process event with process net.exe, parent proce...	ws01	ws01	net.exe	—	—	—
   	Aug 5, 2026 @ 18:17:06.018	APT29 - System and AD Discovery	medi...	47		process event with process reg.exe, parent proce...	ws01	ws01	reg.exe	—	—	—
   	Aug 5, 2026 @ 18:17:06.010	APT29 - System and AD Discovery	medi...	47		process event with process systeminfo.exe, pare...	ws01	ws01	systeminfo.exe	—	—	—
   	Aug 5, 2026 @ 18:16:07.727	APT29 - Obfuscated PowerShell Execution	high	73		process event with process powershell.exe, pare...	ws01	ws01	powershell.exe	—	—	—
   	Aug 5, 2026 @ 18:16:07.714	APT29 - Obfuscated PowerShell Execution	high	73		process event with process cmd.exe, parent proc...	ws01	ws01	cmd.exe	—	—	—
   	Aug 5, 2026 @ 18:15:07.056	APT29 - File Collection via PowerShell	medi...	50		process event with process powershell.exe, pare...	ws01	ws01	powershell.exe	—	—	—
   	Aug 5, 2026 @ 18:14:55.749	APT29 - PowerShell Suspicious Network Outbound	critical	99		process event with process powershell.exe, pare...	ws01	ws01	powershell.exe	—	—	—
   	Aug 5, 2026 @ 18:14:48.724	APT29 - Phishing via HTA and Mshta Proxy Execution	critical	99		process event with process powershell.exe, pare...	ws01	ws01	powershell.exe	—	—	—

Key features demonstrated:

✓ Real-time detection

✓ Honeypot trap fires on

✓ Full technique traceability

Evaluation Metrics

Five SOC metrics evaluated using measured test results and procedure-based estimates.

Metric	Result	Evidence Status
MTTD	18.4 seconds (measured)	Measured
MTTC	~2.1 minutes (estimated)	Procedure-based estimate
MTTR	~5.6 minutes (estimated)	Procedure-based estimate
Technique Coverage	11 / 11 = 100% (measured)	Complete within selected scope
False-Positive Rate	~8.5% (estimated)	Preliminary estimate



Overall verdict: Detection engineering performed strongly within the project scope, with MTTD and MITRE ATT&CK coverage measured from the test results, while MTTC, MTTR, and FPR were estimated using the documented procedures and available evidence.

Key Lessons and Future Improvements

Behavior-Based Detection

LOTL activity becomes visible when process lineage, command execution, registry modifications, and network connections are correlated.

Telemetry Quality

Sysmon and PowerShell Script Block Logging (Event 4104) were absolutely critical to uncovering hidden adversary intent.

Continuous Rule Tuning

Tuning false positives was a natural part of the engineering process, not a failure of the initial rule definitions.

Thank You!

Questions & Discussion

